RACE: Scalable Statistical Estimation of Functional Consistency in LLM Neurons

Runyu Wang[1], Bo Liu[2], Xiaxin Zhang[1], Yu Han[1], Jiawei Cao[1], Xiaoye Zhang[3], Zhe Zhang[4], Yifan Yang[4], Peng Ping[1,*]

[1]Nantong University [2]Chongqing University of Post and Telecommunications
[3]China Southern Power Grid Company Limited [4]Meituan

Stable residual alignment reveals neurons for **specific model behaviors.**

01 Collect Evidence from Residual Stream

Existing attribution approaches face a severe trade-off between computational tractability and functional validity. On one hand, gradient-based methods require iterative backward passes that incur prohibitive overhead, limiting scalable domain-wide auditing. On the other hand, raw activations fail to measure functional consistency due to three key flaws: (1) Lack of direction; (2) Orthogonal cancellation; and (3) Masked instability.

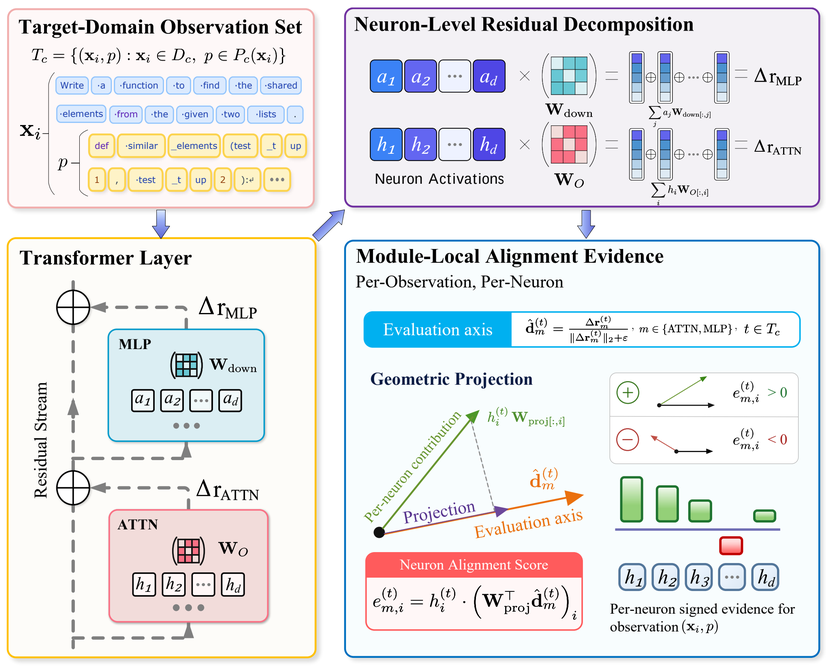


Figure 1: Residual-Direction Alignment (RDA) at forward-pass cost. Each module update is decomposed into rank-1 neuron writes and projected onto the realized update direction $\hat{d}$. The resulting scalar e_j provides signed, module-local evidence: positive values reinforce the module's update, while cancelling orthogonal components are discarded.

02 Estimate, Rank, and Filter

Goal: Evaluating Functional Consistency

A domain-specific neuron must satisfy two desiderata across the observation set T_c:

- **Magnitude:** $\mathbb{E}_{t\in T_c}[e_j^{(t)}]$ is large and positive relative to peers in the module.
- **Stability:** $\mathrm{Var}_{t\in T_c}[e_j^{(t)}]$ is small—not driven by rare outliers or sign fluctuations.

Estimate | Bayesian Evidence Aggregation

Motivation: Simple empirical averages discard cross-input variation and obscure sample-to-sample stability patterns.
NIG Working Model: Treat signed RDA evidence as noisy draws $e_j^{(t)} \sim \mathcal{N}(\mu_j, \sigma_j^2)$ with a conjugate Normal-Inverse-Gamma prior $\mathrm{NIG}(\mu_0, \lambda_0, \alpha_0, \beta_0)$.
O(1) Streaming Inference: Maintaining empirical mean $\bar{e}_j$ and centered sum of squares SS_j yields a closed-form marginal Student's t-posterior for the mean evidence:

$$\mu_j \mid \{e_j^{(t)}\} \sim t_{2\alpha_n}(\mu_{n,j},\ \sigma_{\mu,j}), \qquad \text{where} \quad \sigma^2_{\mu,j} = \frac{\beta_{n,j}}{\alpha_n \lambda_n}.$$

The scale $\sigma_{\mu,j}$ captures epistemic uncertainty—inflated by scarce, noisy, or sign-fluctuating evidence.

Rank | Conservative Alignment Magnitude (CAM)

Motivation: Weak or sparse activations can artificially inflate empirical SNR, while global sorting unfairly favors late layers where updates align closer to logits.
Credible Lower Bound: CAM scores neurons by a one-sided conservative $(1-\gamma)$-lower credible bound on positive mean evidence:

$$\mathrm{CAM}_j(\gamma) = \mathbf{1}[\mu_{n,j} > 0] \cdot \max(0,\ \mu_{n,j} - t_{1-\gamma,2\alpha_n} \cdot \sigma_{\mu,j})\,.$$

Stratified Budget: Neurons are ranked by CAM within each layer and module to preserve coverage across the model's hierarchical computation.

Filter | Reference-Set Filtering (RSF)

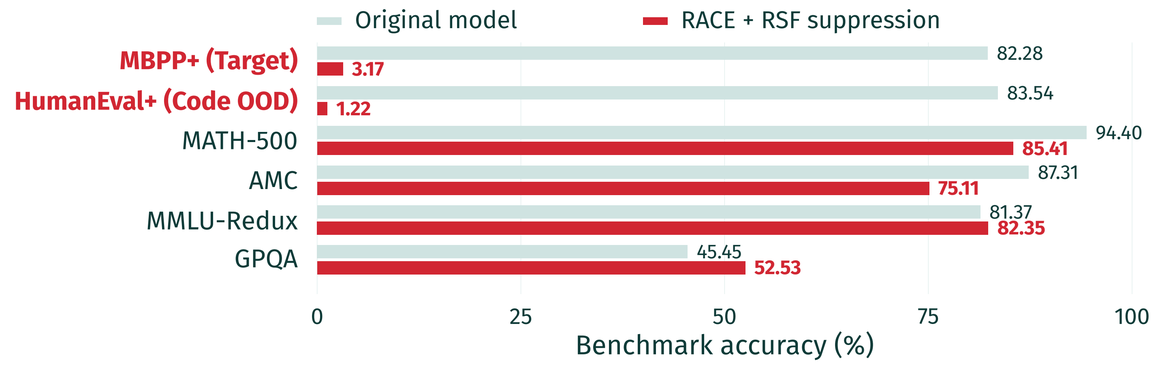
Motivation: Under feature superposition and polysemanticity, raw target rankings conflate domain-specific functions with broadly shared general language capabilities.
Dual-Corpus Filtering: Traverses target ranking Π^ℓ_{tar} and discards neurons that also rank highly on a general reference corpus (WikiText-2, B^ℓ_{ref}) to retain K_{sel} domain neurons:

$$\Pi^\ell_{\mathrm{RSF}} = \left[j \in \Pi^\ell_{\mathrm{tar}} : j \notin B^\ell_{\mathrm{ref}}\right], \qquad S^\ell_{\mathrm{spec}} = \mathrm{First}_{K_{\mathrm{sel}}}(\Pi^\ell_{\mathrm{RSF}}).$$

Prevents baseline linguistic collapse during subsequent behavioral interventions.

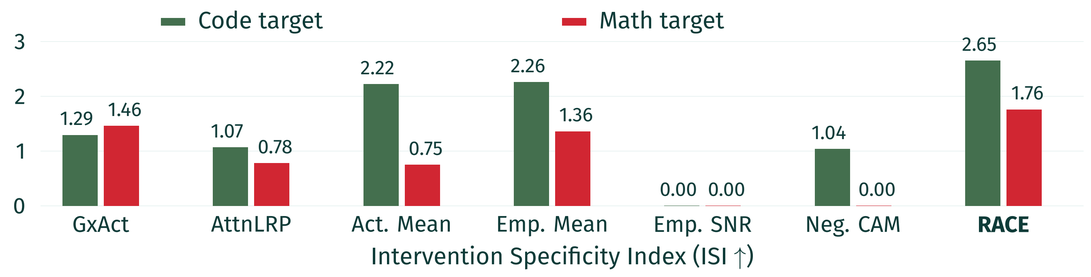
03 Targeted Suppression: Broad to Narrow

Domain-level precision: suppressing the top-1% code-selected MLP neurons collapses the target (**MBPP+**, -96%) and same-domain OOD (**HumanEval+**, -99%) benchmarks, with small effects elsewhere.



Qwen3-4B-it, 3-run average; neurons ranked on MBPP+ with WikiText-2 RSF. Red labels: target and OOD code benchmarks.

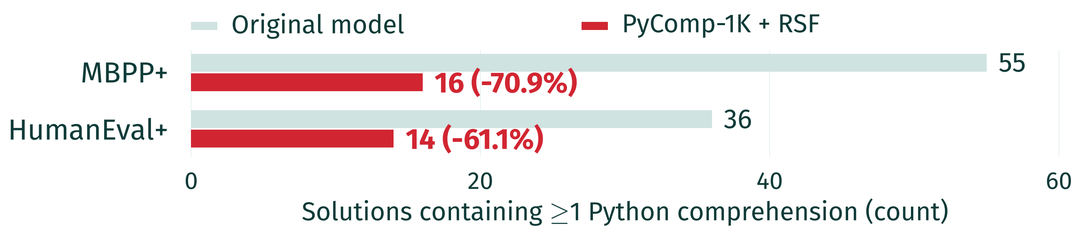
Highest Specificity at <1% of Gradient Cost



Qwen3-4B-it, top-1% MLP budget with RSF; scoring costs 0.63× vs. ∼144× forward FLOPs for GxAct/AttnLRP. ISI $= \max\{0, \log(\Delta_T/(\Delta_G+\delta))\}$, where Δ_T and Δ_G are the average relative accuracy drops on target-domain (incl. OOD) and general benchmarks.

Fine-Grained Steering: Suppressing a Specific Coding Style

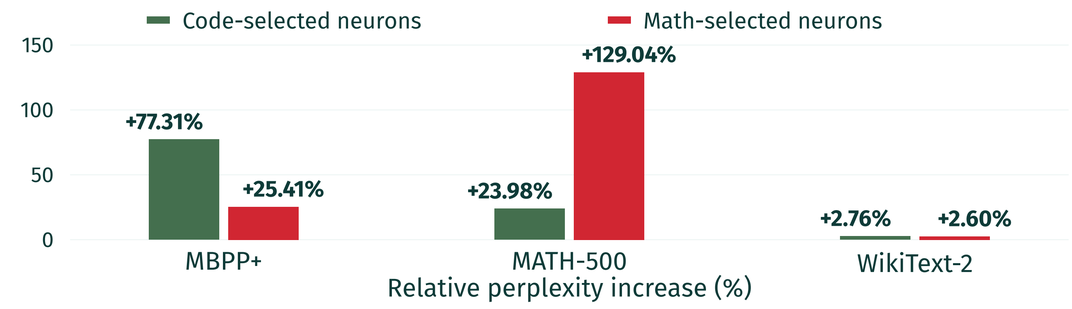
PyComp-1K (1,000 AST-verified comprehensions): suppressing $K_{\mathrm{sel}} = 10$ neurons per layer after RSF cuts comprehension usage by **61%–71%**.



Style shifts, function stays: accuracy on remaining comprehension-using solutions is preserved (MBPP+ 92.73% → 87.50%; HumanEval+ 86.11% → 85.71%).

04 Distributional Verification and Scaling to 32B

Asymmetric shift: perplexity rises sharply on the matched target corpus (**+77.31%** MBPP+, **+129.04%** MATH-500) but changes by less than 3% on WikiText-2.



Qwen3-4B-it, top-1% MLP, WikiText-2 RSF. Green/red: MBPP+/MATH-500 scoring.

Scalability: Consistent Selectivity on 32B Models

OLMo-3.1-32B-it, math scoring + RSF, top-5% MLP: math drops by **>90%**.

Benchmark	Original	Suppressed	Rel. Δ
MATH-500 (target)	86.40	**6.72**	**−92.2%**
AMC (OOD)	79.85	**7.46**	**−90.7%**
GPQA	48.60	37.79	−22.2%
MMLU-Redux	84.70	77.79	−8.2%

Paper, Code, and Contact

Paper Code

Paper: arXiv:2608.24758v2 **Code:** github.com/Nexround/RACE
Contact: 2430310032@stmail.ntu.edu.cn *pingpeng@ntu.edu.cn